JAGADEESHWARAN M

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PROFILE

Passionate and curious AIML engineering student with hands-on experience architecting machine learning systems across computer vision, geospatial analytics, and edge AI. Skilled in Python-based ML pipelines, data analysis, and deploying AI models into real-world applications. Actively participates in hackathons such as the Amazon ML Hackathon and Kaggle competitions to explore practical machine learning challenges. Continuously learning through projects and open-source collaboration.

PROJECTS

FISHVISION : Built an offline-first mobile app to identify ~30 Indian fish species using YOLOv8 + MobileNetV3. Developed a React Native frontend with fast local search and filtering based on habitat, region, and family. Designed a scalable Node.js backend with JWT authentication, SQLite storage and image-processing pipeline for future expansion to 2,500+ species. Runs completely offline.
Tech Stack : YOLOv8, MobileNetV3, Python ,OpenCV, React Native, Expo, Node.js, Express.js, JWT, SQLite.

NEER : AI-Powered Lake Health Monitoring System. Designed an end-to-end geospatial AI system to assess water health of Coimbatore lakes using satellite data (NDWI, NDCI, FAI, MCI, turbidity, thermal). Built time-series regression models to predict BOD, DO, and COD using multi-year Sentinel-2 data. Classified pollution causes and suggested improvement strategies. Visualized results on interactive maps with dynamic filters
Tech Stack : Google Earth Engine, Python, Scikit-learn, Remote Sensing Indices, GeoJSON, React, Flask, leaflet.

ROTUS : Built a real-time object tracking system achieving around 50 fps using Raspberry Pi 5, Google Coral TPU, and ZED 2i camera. Implemented a computer vision pipeline for hardware-accelerated detection with SORT-based multi-object tracking, and stereo disparity-based depth estimation for 3D localization for robotics and edge-AI applications.
Tech Stack : TensorFlow Lite, SSD MobileNet V2, OpenCV, Python, SORT, Linux (Raspberry Pi OS).

EDUCATION

ABC Matriculation Higher Secondary School, Coimbatore, Tamilnadu
2022 - 2023 (HSC) Percentage: 82%

Bachelor of Technology in Artificial Intelligence and Machine Learning
Sri Shakthi Institute of Engineering and Technology, Coimbatore, Tamil Nadu
2023 - 2027 CGPA: 7.41/10

SKILLS

- **Languages:** Python, SQL, C, HTML, CSS, JavaScript
- **Tools/Frameworks:** Flask, FastAPI, Streamlit, Chainlit, Git, Linux, MySQL, Docker
- **AI/ML:** Scikit-learn, PyTorch, LangChain, LangGraph, deepagents, Langsmith, litellm, Pandas, NumPy, Matplotlib, OpenCV, TensorFlow(Keras), TFLite, Ultralytics(YOLO), MobileNet, Ollama, MLflow

CERTIFICATIONS

- Prompt Engineering **by NVIDIA**
- Data Analytics in Python **by NPTEL**
- Machine learning specialization **by Stanford**
- Deep learning specialization **by Deeplearning.AI**